

# Diamonds

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## Part 1

### 1.1 Random Sample Selected

2 categorical variables: cut, color

3 independent quantities: carat, x(width), depth

### 1.2 Data Description and Descriptive Statistics

```
diamonds <- read.csv("Diamonds Prices2022.csv")
head(diamonds)
```

```
##   X carat      cut color clarity depth table price    x    y    z
## 1 1  0.23    Ideal     E    SI2   61.5    55   326 3.95 3.98 2.43
## 2 2  0.21  Premium     E    SI1   59.8    61   326 3.89 3.84 2.31
## 3 3  0.23     Good     E    VS1   56.9    65   327 4.05 4.07 2.31
## 4 4  0.29  Premium     I    VS2   62.4    58   334 4.20 4.23 2.63
## 5 5  0.31     Good     J    SI2   63.3    58   335 4.34 4.35 2.75
## 6 6  0.24 Very Good     J   VVS2   62.8    57   336 3.94 3.96 2.48
```

```
set.seed(100)
sample_data <- diamonds[sample(nrow(diamonds), 600), ]
summary(sample_data)
```

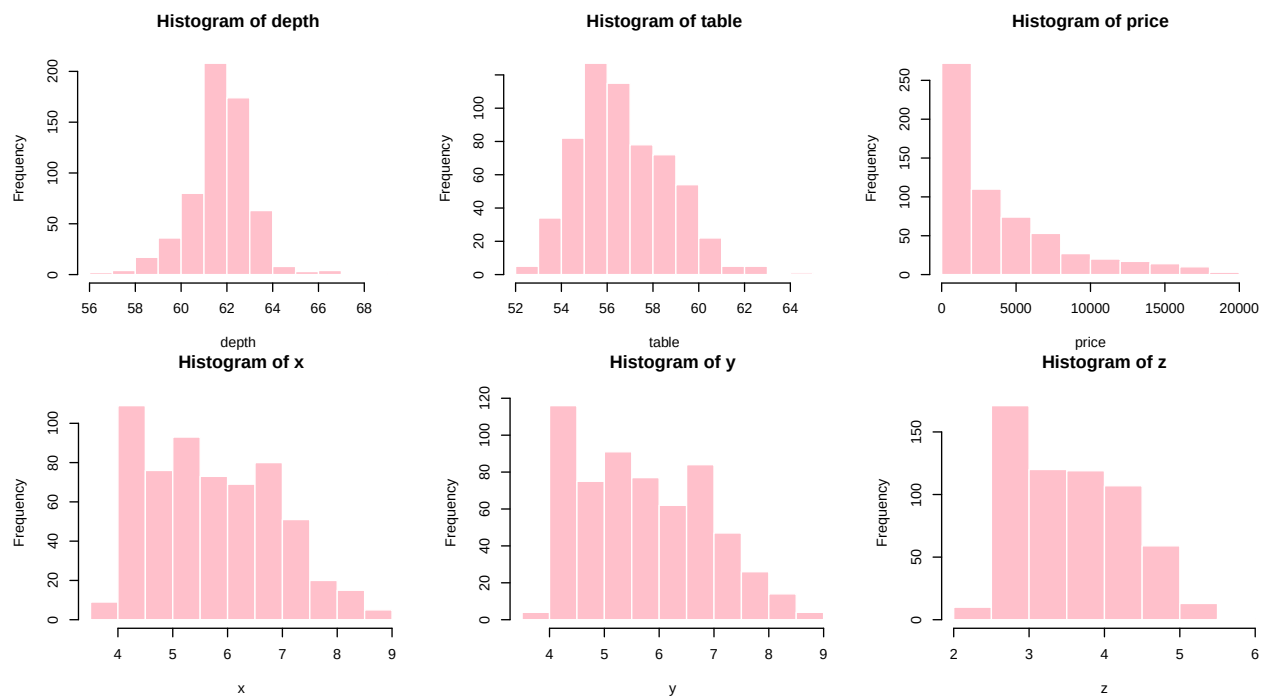
```
##           X           carat           cut           color
##  Min.      : 26   Min.    :0.2300   Length:600   Length:600
## 1st Qu.:15707 1st Qu.:0.4000   Class :character   Class :character
## Median :27331 Median :0.7000   Mode  :character   Mode  :character
## Mean    :27951 Mean    :0.8019
## 3rd Qu.:41451 3rd Qu.:1.0825
## Max.    :53931 Max.    :2.5400
## clarity      depth           table           price
## Length:600   Min.      :56.20   Min.      :52.80   Min.      : 354.0
## Class :character 1st Qu.:61.10   1st Qu.:56.00   1st Qu.: 981.2
## Mode  :character Median :61.90   Median :57.00   Median : 2304.5
```

```
##           Mean   :61.77   Mean   :57.19   Mean   : 4029.1
##           3rd Qu.:62.50   3rd Qu.:59.00   3rd Qu.: 5608.5
##           Max.   :68.40   Max.   :65.00   Max.   :18678.0
##           x           y           z
## Min.    :3.940   Min.    :3.900   Min.    :2.360
## 1st Qu.:4.710   1st Qu.:4.710   1st Qu.:2.910
## Median :5.645   Median :5.670   Median :3.500
## Mean    :5.736   Mean    :5.738   Mean    :3.543
## 3rd Qu.:6.640   3rd Qu.:6.605   3rd Qu.:4.072
## Max.    :8.820   Max.    :8.760   Max.    :5.550
```

```
continuous_vars <- c("depth", "table", "price", "x", "y", "z")

par(mfrow = c(1, 3))

for (v in continuous_vars) {
  hist(sample_data[[v]],
    main = paste("Histogram of", v),
    xlab = v,
    col = "pink",
    border = "white")
}
```



**Depth:** The distribution shape is approximately symmetric and bell-shaped. The mean is around 62-63 and doesn't have high variance. There isn't a extreme skewness, so it is a normal distribution.

**Table:** Slightly right-skewed, with most values around 54-58.

**Price:** Strongly right-skewed. Most diamonds in the sample have relatively low prices, only a small number of expensive diamonds create a long right tail.

**x:** Slightly right-skewed, range is between 3-9 and many diamonds fall between 4-6 mm in length.

**y:** Similar to x, the distribution is right-skewed.

**z:** Also right-skewed, with most observations between 2.8-4. A few much larger values extend the right tail.

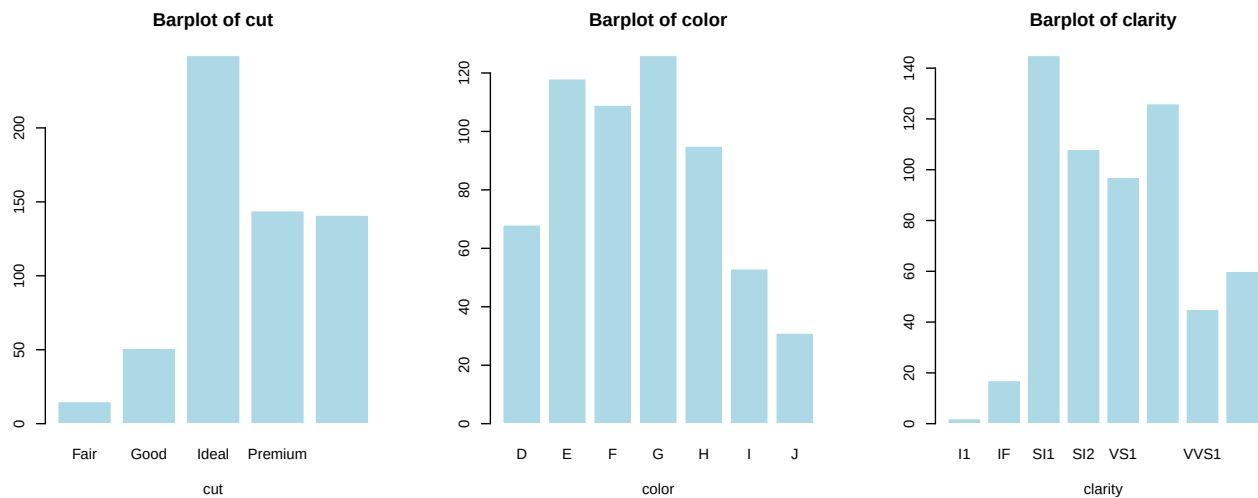
```

categorical_vars <- c("cut", "color", "clarity")

par(mfrow = c(1, 3))

for (v in categorical_vars) {
  barplot(table(sample_data[[v]]),
    main = paste("Barplot of", v),
    xlab = v,
    col = "lightblue",
    border = "white")
}

```



```

par(mfrow = c(1, 1))

```

## 1.3 Determine Correlation

```

library(GGally)

```

```

## Loading required package: ggplot2

```

```

## Warning: package 'ggplot2' was built under R version 4.5.2

```

```

##

```

```

## Attaching package: 'ggplot2'

```

```

## The following object is masked _by_ '.GlobalEnv':

```

```

##

```

```

## diamonds

```

```

library(dplyr)

```

```

##

```

```

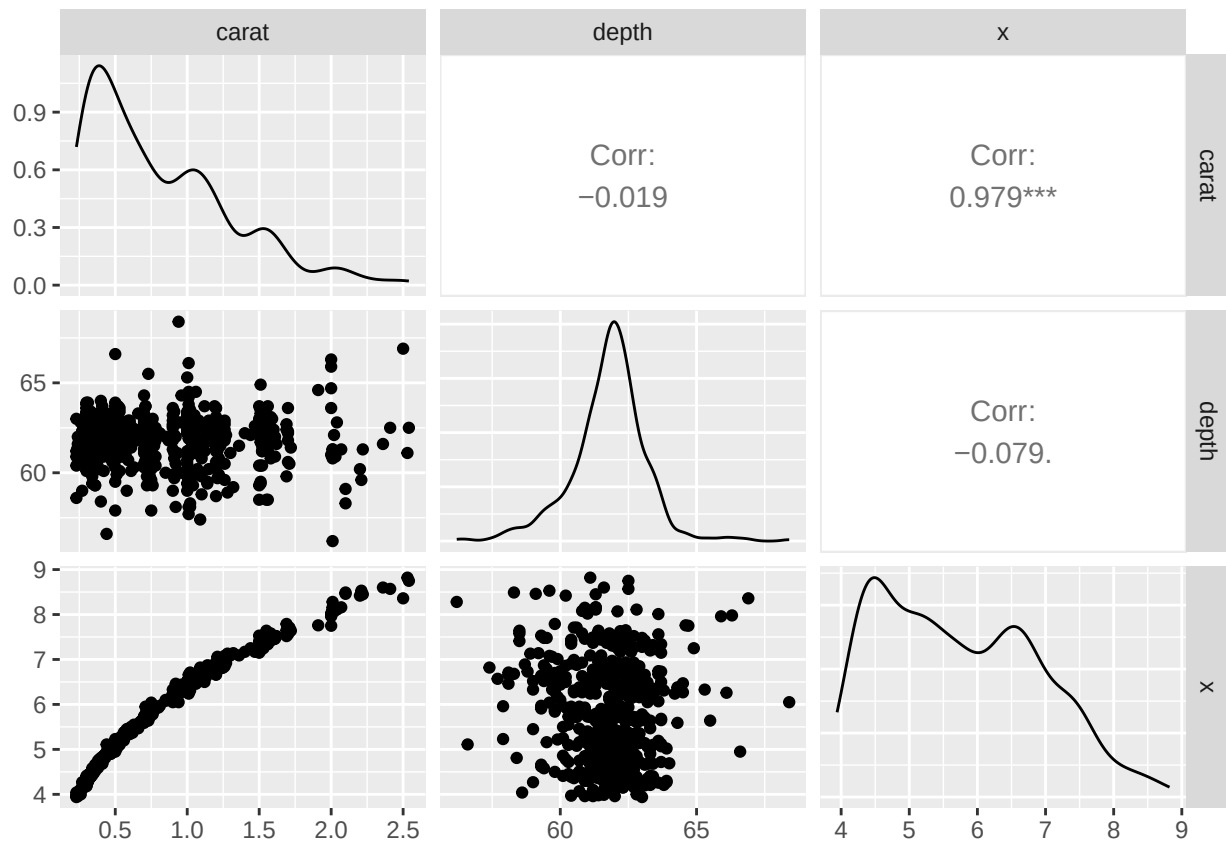
## Attaching package: 'dplyr'

```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
sample_data %>%
  dplyr::select(carat, depth, x) %>%
  ggpairs(progress = F)
```



**carat vs. depth** The correlation between carat and depth is extremely weak ( $r = -0.019$ ), indicating essentially no linear relationship between these two variables.

**Carat vs. x** Carat and x exhibit an extremely strong positive correlation ( $r = 0.979$ ). This indicates that as the carat weight increases, the x dimension (length) of the diamond increases. The strong correlation means there is a multicollinearity, meaning these variables should not be included together in the regression model.

**Depth vs. x** Depth and x show a weak negative correlation ( $r = -0.079$ ), indicating essentially no linear relationship between these two variables.

```
tbl_cut_color <- table(sample_data$cut, sample_data$color)
tbl_cut_color
```

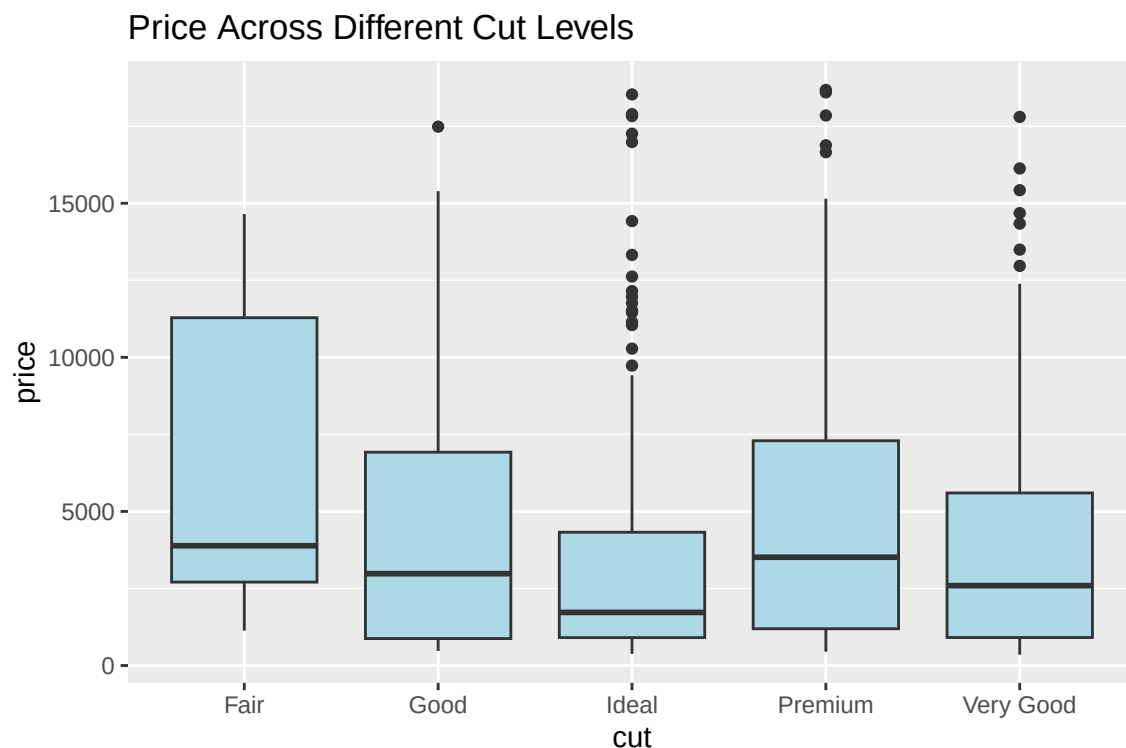
```
##
##           D  E  F  G  H  I  J
## Fair      1  0  1  3  4  3  3
## Good      8  8  9  9  4  9  4
## Ideal     31 44 55 55 33 23  8
## Premium   14 28 28 35 28  4  7
## Very Good 14 38 16 24 26 14  9
```

```
chisq.test(tbl_cut_color)
```

```
## Warning in chisq.test(tbl_cut_color): Chi-squared approximation may be
## incorrect
```

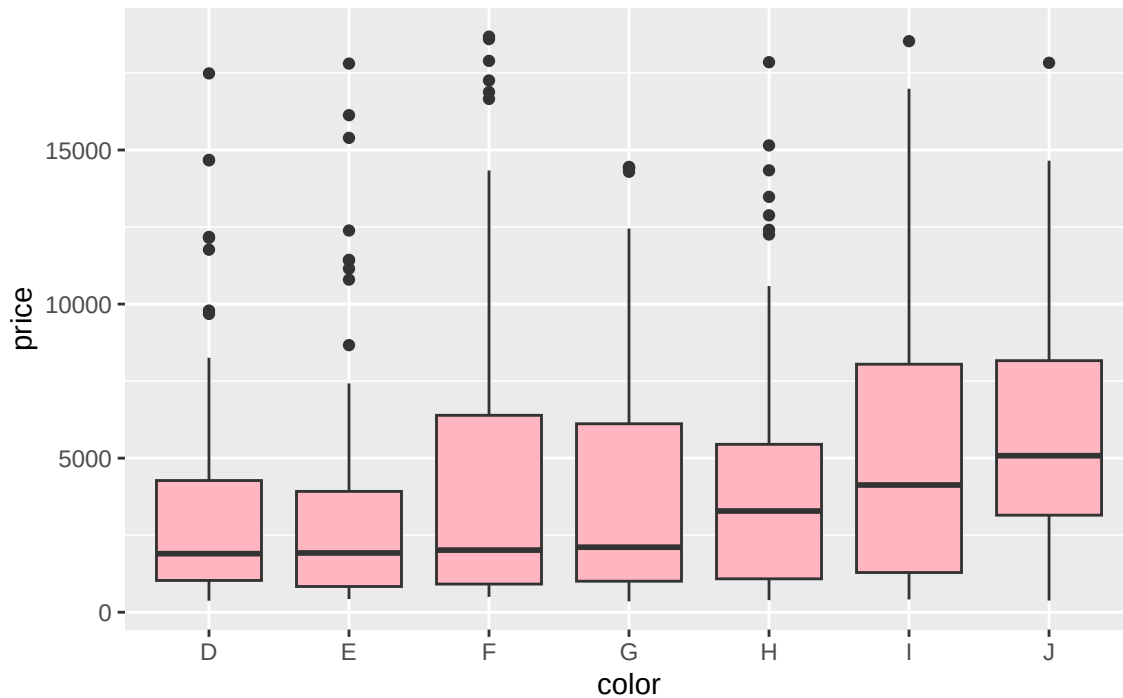
```
##
## Pearson's Chi-squared test
##
## data:  tbl_cut_color
## X-squared = 47.03, df = 24, p-value = 0.003322
```

```
par(mfrow = c(1, 3))
ggplot(sample_data, aes(x = cut, y = price)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "Price Across Different Cut Levels")
```



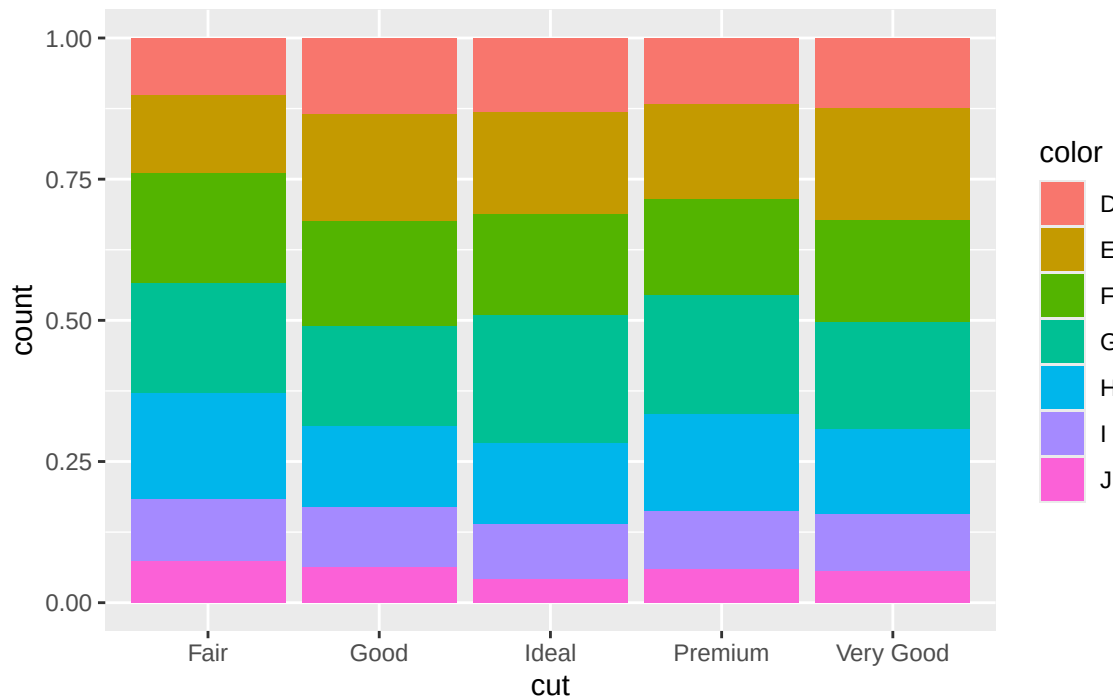
```
ggplot(sample_data, aes(x = color, y = price)) +
  geom_boxplot(fill = "lightpink") +
  labs(title = "Price Across Different Color Grades")
```

Price Across Different Color Grades



```
ggplot(diamonds, aes(x = cut, fill = color)) +  
  geom_bar(position = "fill") +  
  labs(title = "Cut Across Different Color Grades")
```

Cut Across Different Color Grades



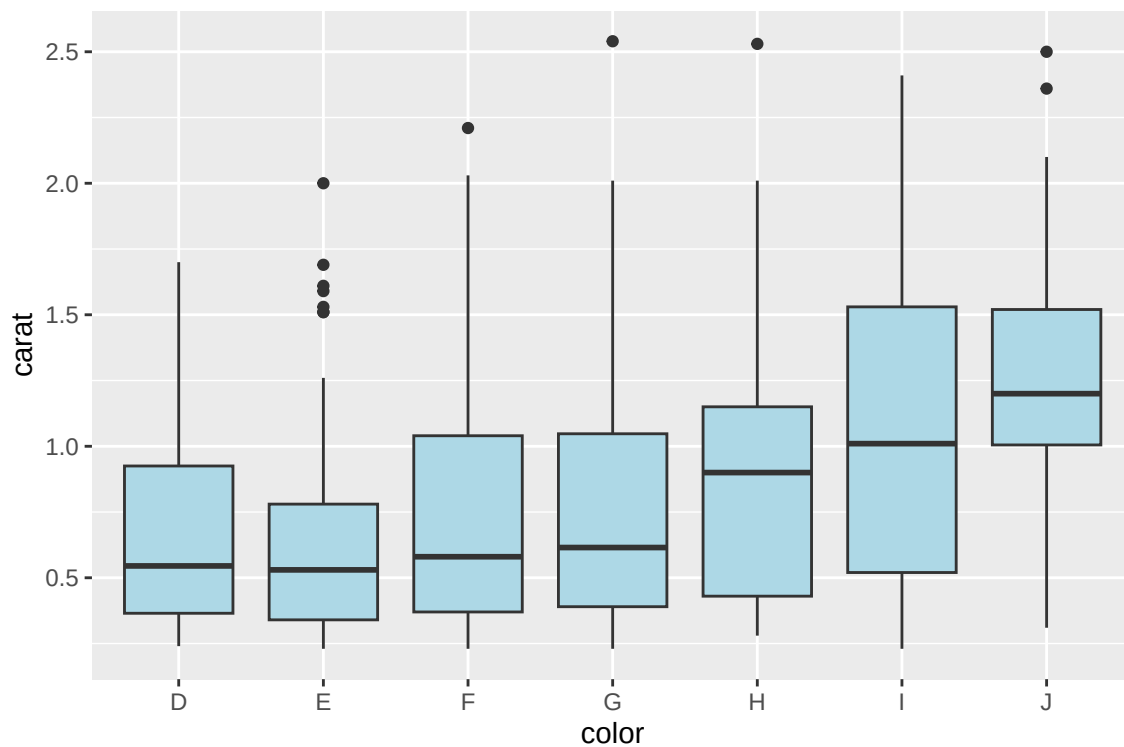
**price vs. cut** From the relationship between price level and different cut levels, we can conclude that different price level intend to have different price. Fair cutting level's price mean is only a bit higher than others, and other cutting level have similar price mean. We can't see an obvious correlation between Cut

Levels and Price.

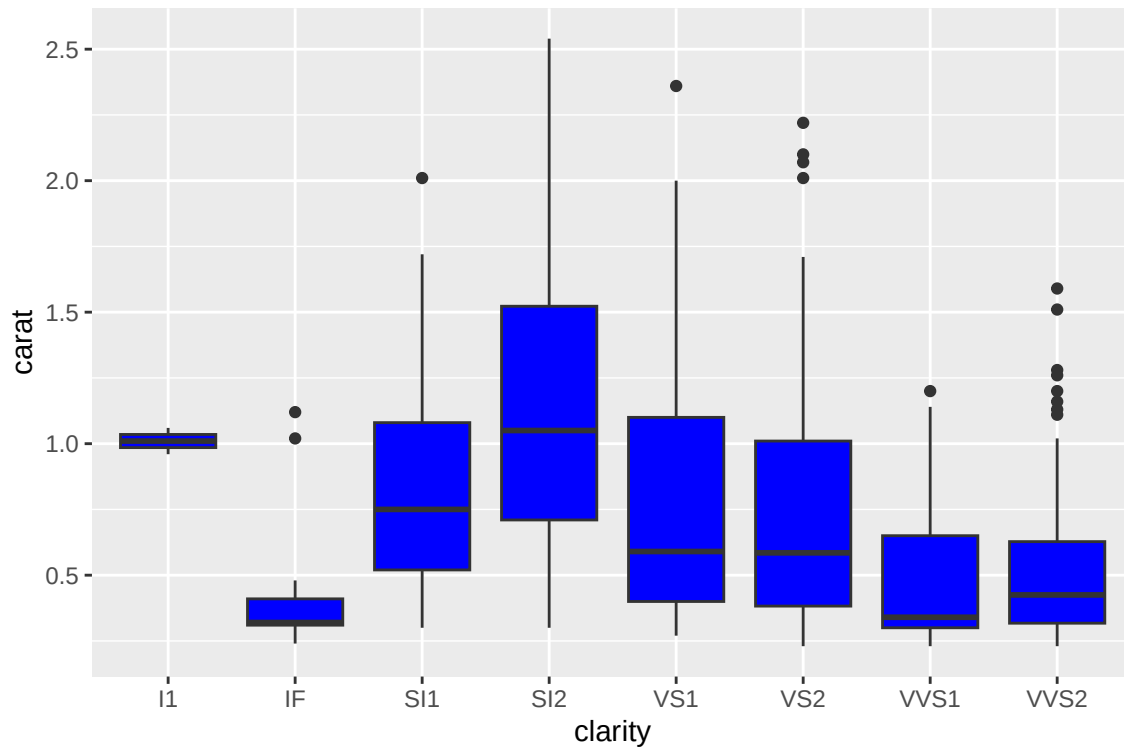
**price vs. color** From the graph, we can see there is a slight positive correlation between price and color. As D has the lowest price mean, and J has the highest price mean. So when the color of diamond has the color that is closer to J, it has a higher price.

**cut vs. color** The stacked bar plot shows that the proportions of color are almost uniform across cut levels. This suggests that cut and color do not have a noticeable association.

```
ggplot(sample_data, aes(x = color, y = carat)) +  
  geom_boxplot(fill = "lightblue")
```



```
ggplot(sample_data, aes(x = clarity, y = carat)) +  
  geom_boxplot(fill = "blue")
```

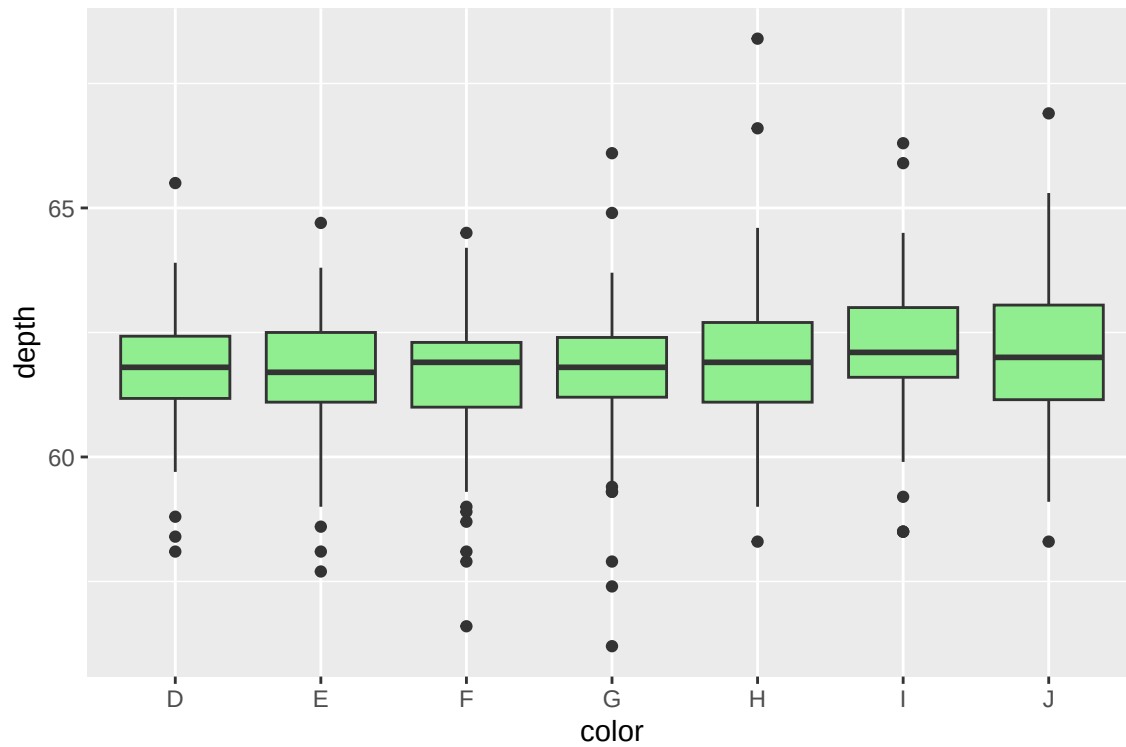


**color-vs-carat** The boxplot of carat across different color grades shows a clear trend that diamonds with lower color quality (I-J) tend to have larger carat values compared to higher color grades (D-F). This indicates a negative association between color grade and carat size.

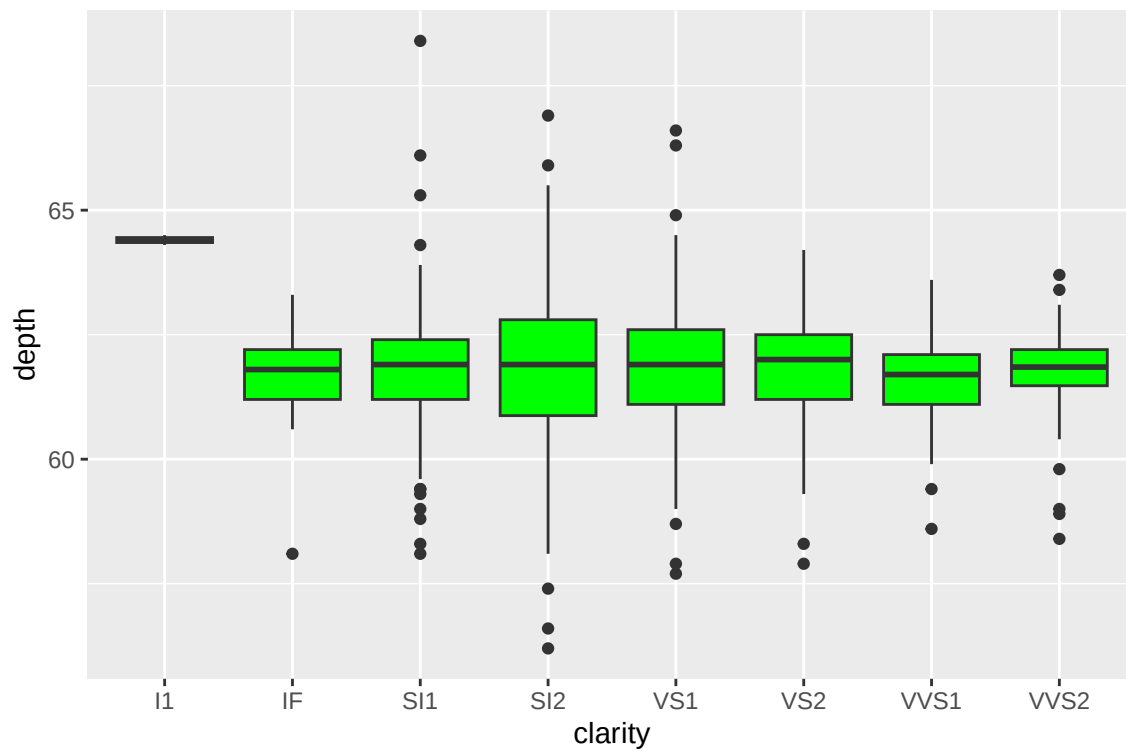
**clarity-vs-carat** The boxplot of carat across clarity levels shows a decreasing trend in median carat from lower clarity grades (I1, SI2) to higher clarity grades (VS, VVS). This means that clarity and carat are negatively associated.

```
ggplot(sample_data, aes(x = color, y = depth)) +  
  geom_boxplot(fill = "lightgreen")
```





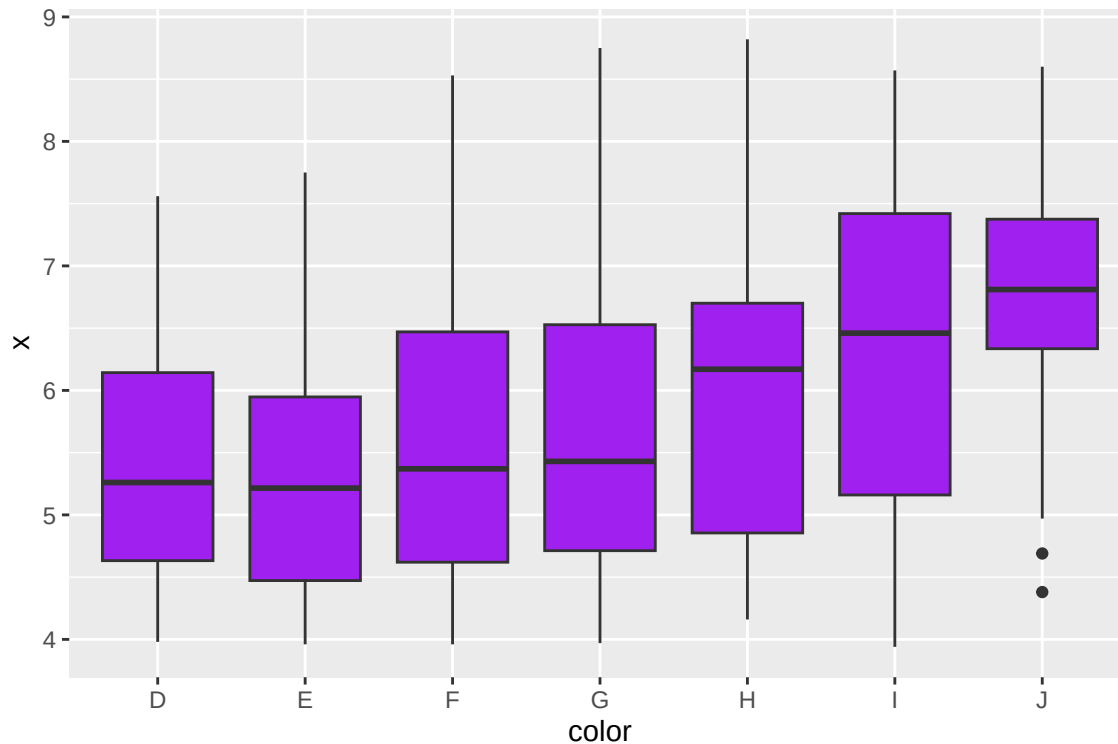
```
ggplot(sample_data, aes(x = clarity, y = depth)) +
  geom_boxplot(fill = "green")
```



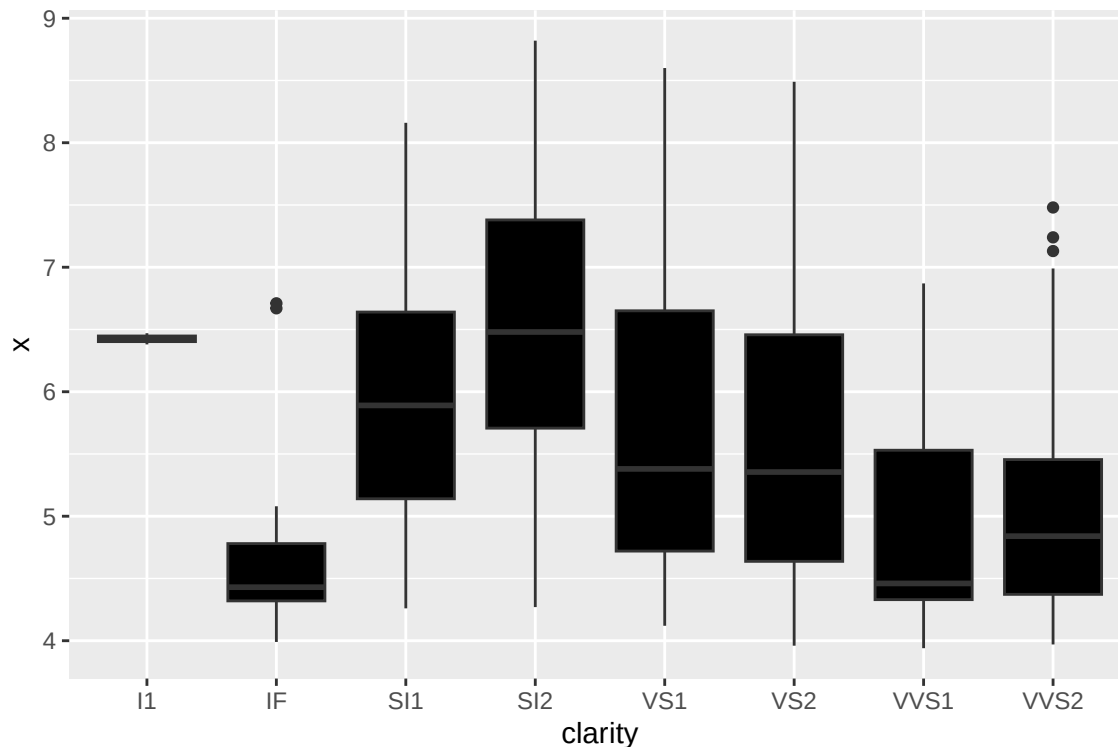
**color-vs-depth** The boxplot comparing depth across different color grades (D–J) shows that depth does not vary substantially across color categories. So depth is not strongly associated with color

**clarity-vs-depth** The boxplot of depth across clarity levels similarly shows only small differences between categories. So depth is not strongly associated with clarity.

```
ggplot(sample_data, aes(x = color, y = x)) +  
  geom_boxplot(fill = "purple")
```



```
ggplot(sample_data, aes(x = clarity, y = x)) +  
  geom_boxplot(fill = "black")
```



**color-vs-x** The boxplot shows a clear pattern in which the x dimension of the diamond increases as the color grade decreases from D to J. The increasing trend indicates a positive association between color category (toward lower grades) and x dimension, meaning that diamonds with worse color grades tend to have larger x values.

**clarity-vs-x** The boxplot comparing x across clarity categories shows more variability but still reveals noticeable patterns. The relationship indicates that x tends to increase as clarity grade decreases.

## 1.4 Multiple Linear Regression Model

```
model_all <- lm(price ~ carat + depth + cut + color, data = sample_data)
summary(model_all)
```

```
##
## Call:
## lm(formula = price ~ carat + depth + cut + color, data = sample_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4515.5  -780.0   -95.1    522.7   6448.9
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -3156.867    3159.831  -0.999  0.318176
## carat         8369.042     133.758   62.569 < 2e-16 ***
## depth        -6.431       48.278   -0.133  0.894080
## cutGood       1214.779     440.194    2.760  0.005967 **
```

```
## cutIdeal      1591.749    417.385    3.814 0.000151 ***
## cutPremium    1412.234    430.883    3.278 0.001109 **
## cutVery Good  1569.313    423.892    3.702 0.000234 ***
## colorE        -292.558    220.058   -1.329 0.184214
## colorF         56.052    223.306    0.251 0.801895
## colorG        -430.946    217.142   -1.985 0.047651 *
## colorH       -1265.109    231.403   -5.467 6.77e-08 ***
## colorI       -1552.215    270.120   -5.746 1.47e-08 ***
## colorJ       -2404.716    323.310   -7.438 3.65e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1439 on 587 degrees of freedom
## Multiple R-squared:  0.8801, Adjusted R-squared:  0.8777
## F-statistic: 359.1 on 12 and 587 DF,  p-value: < 2.2e-16
```

## 1.5 Comment

**carat** As we expect, carat variable has significant to price. It's estimated value is 8369.04 and p-value less than 2e-16 all shows carat is the main effect of diamond price.

**Color** Color variable is the one that surprise me which five out of 6 color's estimated value are negative. I expected that color may increase diamond price, and some of the color may significant increase price.

## Part 2

### 2.1 simple linear regression of carat and price

```
model_simple <- lm(price ~ carat, data = sample_data)
summary(model_simple)
```

```
##
## Call:
## lm(formula = price ~ carat, data = sample_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6063.5  -861.2   -61.5   537.2  7544.9
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -2242.6      126.0   -17.80  <2e-16 ***
## carat         7821.0      134.5    58.15  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1596 on 598 degrees of freedom
## Multiple R-squared:  0.8497, Adjusted R-squared:  0.8495
## F-statistic: 3381 on 1 and 598 DF,  p-value: < 2.2e-16
```

## 2.2 Interpret simple regression

The intercept represents the predicted price when carat = 0, which is not meaningful in practice because diamonds cannot have zero carat. For every additional one unit increase in carat, the price increases by approximately 7821 dollars on average.

The average deviation of intercept has 126 stand to its population value. The average deviation of carat would vary by 134.5 across repeated samples to its population value.

The intercept is -17.80 standard errors from 0. The hour studied has 58.15 standard error from 0.

We test  $H_0 : \beta_1 = 0$  and  $H_a : \beta_1 \neq 0$

The p-value of study ( $< 2e-16$ ) is smaller than 0.05, we reject to the null hypothesis. We conclude that carat significantly increase price of diamonds.

F-statistic is 3381, which is extremely large to reject the null hypothesis.

The Adjusted R-squared is 0.8495, indicating that approximately 84.95% of the variance in price can be explained by carat alone. The Adjusted R-squared is closer to 1 and it is closer to Multiple R-squared (0.8497). Carat is a strong predictor of price.

```
confint(model_simple)
```

```
##                2.5 %    97.5 %  
## (Intercept) -2490.106 -1995.131  
## carat       7556.877  8085.182
```

The confident interval of intercept is [-2490.106, -1995.131].

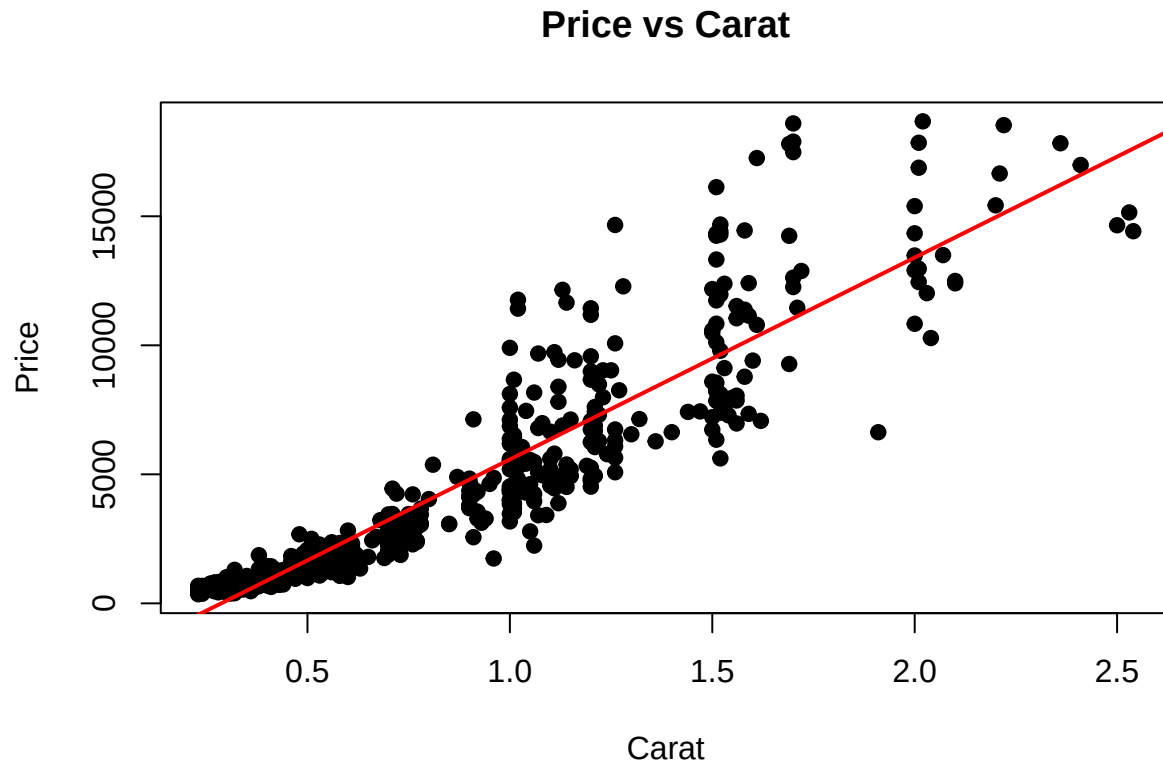
The confident interval of carat is [7556.877, 8085.182].

```
predict(model_simple, newdata = data.frame(carat = 1),  
        interval = "prediction")
```

```
##          fit      lwr      upr  
## 1 5578.411 2440.331 8716.491
```

The prediction interval of carat is [2440.331, 8716.491].

```
plot(sample_data$carat, sample_data$price,  
     main = "Price vs Carat",  
     xlab = "Carat",  
     ylab = "Price",  
     pch = 19, col = "black")  
  
abline(model_simple, col = "red", lwd = 2)
```



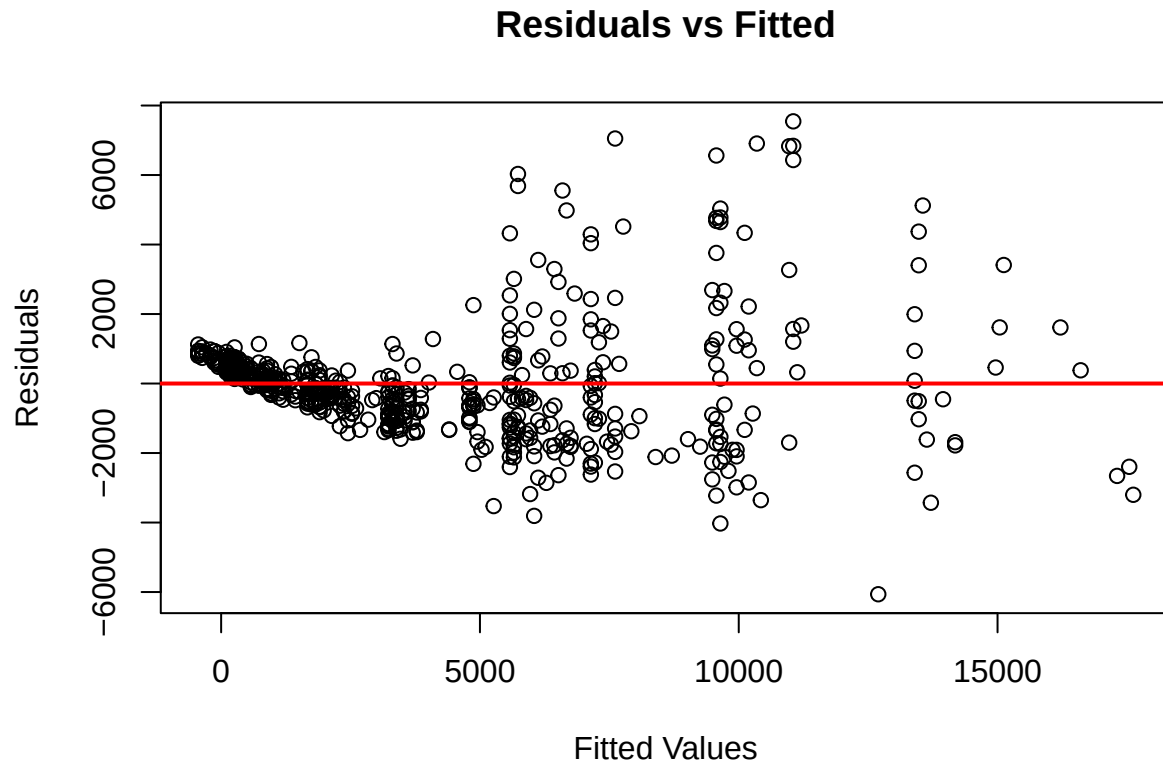
From the scatter plot, we can see that the price and carat has a positive relationship.

## 2.3 Test assumption and transformations

Test assumption:

Check 1: linearity and homoscedasticity

```
plot(model_simple$fitted.values, resid(model_simple),  
     main="Residuals vs Fitted",  
     xlab="Fitted Values", ylab="Residuals")  
abline(h=0, col="red", lwd=2)
```

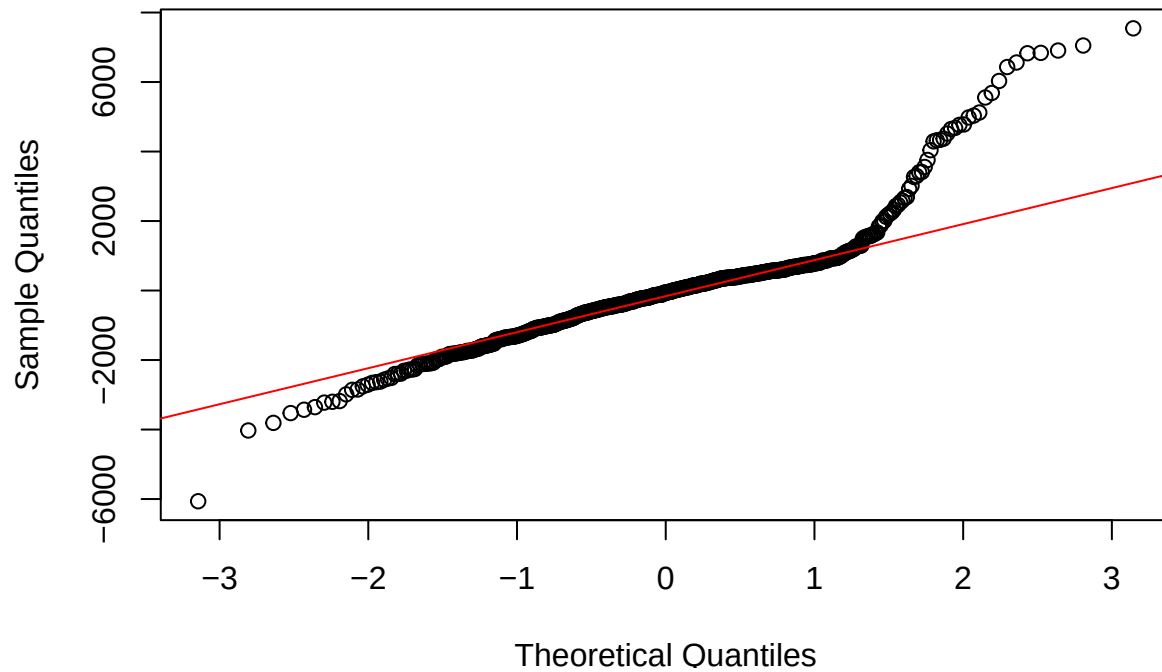


The graph has a curvature structure. This indicates that the relationship between carat and price is not linear on the original scale. The spread of residuals increases substantially as fitted values increase. This funnel-shaped pattern indicates heteroscedasticity. So we suggest the need for transformations.

Check 2: normality

```
qqnorm(resid(model_simple))  
qqline(resid(model_simple), col="red")
```

## Normal Q-Q Plot

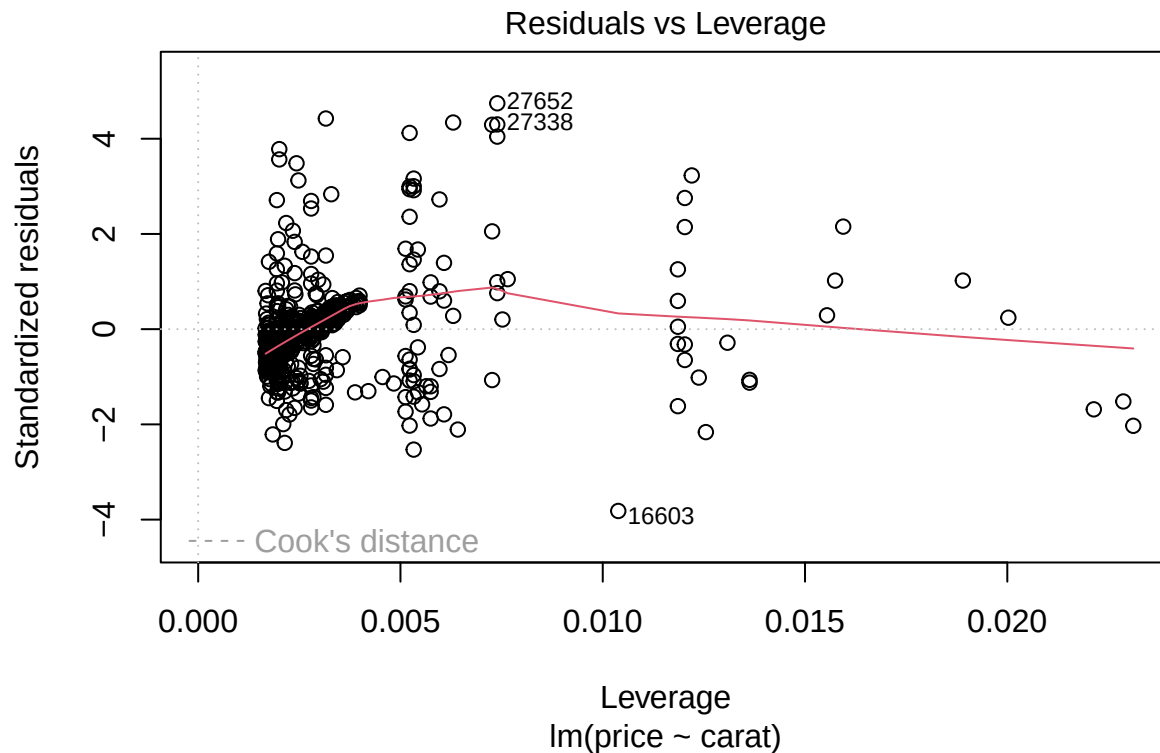


Most residuals follow a normal distribution, but the tails show strong deviations. The residuals are not normally distributed.

Check 3: Residuals vs. Leverage

```
plot(model_simple, which = 5)
```





Leverage measures how far an observation's predictor values are from the mean of the predictor value. From the graph, we see that most observations have low leverage and fall well within the Cook's distance boundaries. But some of them have large standardized residuals that exhibit outliers in the response variable. Also, some observations have high leverage that is from 0.02 to 0.03 or more, meaning they might influence the slope of the regression line. All of the observations are within the Cook's distance boundaries, so no observation is extremely influential. Due to the existing high leverage and outliers, we need to use transformation like  $\log(\text{price})$ .

### Transformation:

```
# log
y <- log(sample_data$price)
model_simple_log <- lm(y ~ carat, data = sample_data)

# x^2
x <- sample_data$carat^2
model_simple_x2 <- lm(price ~ x, data = sample_data)
```

## 2.4 Interpret transformation

```
summary(model_simple_log)
```

```
##
## Call:
## lm(formula = y ~ carat, data = sample_data)
##
## Residuals:
```

```
##      Min      1Q   Median      3Q      Max
## -1.43151 -0.17114  0.05517  0.18371  0.50955
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.88959    0.01885   206.4  <2e-16 ***
## carat       2.30477    0.02012   114.6  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2388 on 598 degrees of freedom
## Multiple R-squared:  0.9564, Adjusted R-squared:  0.9564
## F-statistic: 1.313e+04 on 1 and 598 DF,  p-value: < 2.2e-16
```

```
summary(model_simple_x2)
```

```
##
## Call:
## lm(formula = price ~ x, data = sample_data)
##
## Residuals:
##      Min      1Q   Median      3Q      Max
## -4340.6 -1316.1  -276.1  1050.5  8286.6
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -14220.20    391.37  -36.33  <2e-16 ***
## x           3181.79     66.91   47.56  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1883 on 598 degrees of freedom
## Multiple R-squared:  0.7909, Adjusted R-squared:  0.7905
## F-statistic: 2262 on 1 and 598 DF,  p-value: < 2.2e-16
```

Compared transformation  $\log(\text{price})$  and  $x$  square, we prefer to use  $\log(\text{price})$  since the total variance of it is 95.83% in response variable that is larger than the total variance of model  $x$  square (78.71%).

Compared to the previous summary of simple regression, we find that the estimator and standard error is lowered, meaning the average deviation from its true population value is lowered. Also, the adjusted R-squared increases from 0.8639 to 0.9583, showing that the transformed model explains a much larger proportion of the variance in price. The Multiple R-squared is more closer to adjusted R-squared by 0.0001 difference. Overall, the transformation model makes a better model fit.

## 2.5 Add other variables

```
##
## Call:
## lm(formula = y ~ carat + depth, data = sample_data)
##
## Residuals:
##      Min      1Q   Median      3Q      Max
```

```
## -1.13608 -0.16502 0.06799 0.18274 0.30220
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.390029   0.419989  17.596 < 2e-16 ***
## carat        2.301755   0.019058 120.777 < 2e-16 ***
## depth       -0.056627   0.006788  -8.342 5.07e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2261 on 597 degrees of freedom
## Multiple R-squared:  0.961, Adjusted R-squared:  0.9608
## F-statistic: 7350 on 2 and 597 DF, p-value: < 2.2e-16

##
## Call:
## lm(formula = y ~ carat + depth + x, data = sample_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47722 -0.04144  0.01172  0.03844  0.17138
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.495805   0.124036   3.997 7.21e-05 ***
## carat        0.035895   0.023666   1.517 0.129857
## depth       -0.006339   0.001727  -3.671 0.000263 ***
## x           0.977204   0.010009  97.628 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05491 on 596 degrees of freedom
## Multiple R-squared:  0.9977, Adjusted R-squared:  0.9977
## F-statistic: 8.63e+04 on 3 and 596 DF, p-value: < 2.2e-16

##
## Call:
## lm(formula = y ~ carat + depth + x + color, data = sample_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47166 -0.04130  0.01088  0.03727  0.17109
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.469794   0.124930   3.760 0.000187 ***
## carat        0.029940   0.023920   1.252 0.211190
## depth       -0.006227   0.001736  -3.588 0.000361 ***
## x           0.979431   0.010050  97.455 < 2e-16 ***
## colorE       0.019312   0.008356   2.311 0.021165 *
## colorF       0.010438   0.008507   1.227 0.220299
## colorG       0.010159   0.008260   1.230 0.219253
## colorH       0.004757   0.008774   0.542 0.587894
## colorI       0.017674   0.010265   1.722 0.085644 .
```

```
## colorJ      0.018315   0.012319   1.487 0.137611
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05482 on 590 degrees of freedom
## Multiple R-squared:  0.9977, Adjusted R-squared:  0.9977
## F-statistic: 2.886e+04 on 9 and 590 DF,  p-value: < 2.2e-16

##
## Call:
## lm(formula = y ~ carat + depth + x + color + clarity, data = sample_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.198402 -0.040707  0.009085  0.036288  0.174864
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.014234   0.133009   0.107  0.91481
## carat        0.014571   0.023534   0.619  0.53607
## depth       -0.004428   0.001678  -2.639  0.00854 **
## x           0.987591   0.009953  99.222 < 2e-16 ***
## colorE       0.020249   0.007964   2.543  0.01126 *
## colorF       0.010000   0.008128   1.230  0.21905
## colorG       0.009785   0.007974   1.227  0.22024
## colorH       0.008596   0.008402   1.023  0.30666
## colorI       0.023290   0.009822   2.371  0.01805 *
## colorJ       0.017252   0.011733   1.470  0.14197
## clarityIF    0.326511   0.039824   8.199 1.55e-15 ***
## claritySI1   0.310045   0.037536   8.260 9.83e-16 ***
## claritySI2   0.302796   0.037739   8.023 5.68e-15 ***
## clarityVS1   0.312190   0.037776   8.264 9.51e-16 ***
## clarityVS2   0.309579   0.037718   8.208 1.45e-15 ***
## clarityVVS1  0.310316   0.038441   8.072 3.96e-15 ***
## clarityVVS2  0.314361   0.038178   8.234 1.19e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05204 on 583 degrees of freedom
## Multiple R-squared:  0.998, Adjusted R-squared:  0.9979
## F-statistic: 1.802e+04 on 16 and 583 DF,  p-value: < 2.2e-16
```

The addition of depth and x help to increase the adjusted R-squared. The addition of color and clarity did not change the adjusted R-squared(0.9977), indicating that the variable does not provide additional explanatory power beyond the existing predictors. This suggests that the new variable is not a meaningful contributor to the model.

## 6. Interest that occurred

We are surprised that after transforming model  $x^2$  has a lower adjusted  $R^2 = 0.7871$  than original model adjusted  $R^2 = 0.8639$ . Since  $x^2$  transformation is supposed to increase the skewness and worsen linearity.

Also, the categorical variables (Color and Clarity) do not change the adjusted R-squared. This means these two variables don't improve the goodness of model, and the best model will include carat, depth and x.

## Part 3

### 3.1 BACKWARD AIC

```
library(generics)
```

```
##
## Attaching package: 'generics'

## The following object is masked from 'package:dplyr':
##
##     explain

## The following objects are masked from 'package:base':
##
##     as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
##     setequal, union
```

```
library(broom)
```

```
step(model_simple_log, direction = "backward")
```

```
## Start:  AIC=-3530.17
## y ~ carat + depth + x + color + clarity
##
##           Df Sum of Sq    RSS    AIC
## - color     6    0.0270  1.6058 -3532.0
## - carat     1    0.0010  1.5798 -3531.8
## <none>                        1.5788 -3530.2
## - depth     1    0.0189  1.5976 -3525.0
## - clarity   7    0.1940  1.7728 -3474.6
## - x         1   26.6605 28.2393 -1801.7
##
## Step:  AIC=-3531.99
## y ~ carat + depth + x + clarity
##
##           Df Sum of Sq    RSS    AIC
## - carat     1    0.0023  1.6081 -3533.1
## <none>                        1.6058 -3532.0
## - depth     1    0.0192  1.6250 -3526.9
## - clarity   7    0.1910  1.7968 -3478.6
## - x         1   26.9077 28.5135 -1807.9
##
## Step:  AIC=-3533.12
## y ~ depth + x + clarity
##
##           Df Sum of Sq    RSS    AIC
## <none>                        1.61 -3533.1
## - depth     1     0.02    1.62 -3528.9
## - clarity   7     0.20    1.80 -3478.2
## - x         1   638.59 640.20   56.9
```

```
##
## Call:
## lm(formula = y ~ depth + x + clarity, data = sample_data)
##
## Coefficients:
## (Intercept)      depth           x    clarityIF    claritySI1    claritySI2
##   -0.027330   -0.003955    0.994089    0.326279    0.308665    0.302510
##   clarityVS1   clarityVS2   clarityVVS1   clarityVVS2
##    0.310679    0.307982    0.310422    0.313109
```

After looking at the outcome of AIC function, we can conclude that depth, clarity and x variables create the best model. That is because AIC become smaller after removing color and carat variable, which improve the goodness of model.

```
#document the observation for useful variables
newdata <- sample_data %>%
  select(price, depth, clarity, x)

head(newdata)
```

```
##      price depth clarity    x
## 16887  6738  59.6     SI1 7.08
##  3696   3448  62.7     VS2 5.65
## 31705    770  62.0     SI1 4.59
## 24270 12494  59.1     SI2 8.46
## 11159  4946  59.7     SI2 7.06
## 26116 15393  64.7     SI2 7.75
```

## 3.2 VIF

```
model_final <- lm(y ~ depth + clarity + x, data = newdata)
library(car)
```

```
## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##      recode

vif(model_final)
```

```
##      GVIF Df GVIF^(1/(2*Df))
## depth  1.027833  1      1.013821
## clarity 1.238922  7      1.015421
## x      1.225795  1      1.107156
```

All depth, clarity and x variables' GVIF value are less than 5, which means there isn't significant multicollinearity. We can ignore the multicollinearity in this model, and the prediction is reasonable.

### 3.3 CI and PI

```
predicted_data <- data.frame(  
  clarity = "SI2",  
  depth = 60,  
  x = 4  
)
```

```
#CI  
predict(model_final, newdata = predicted_data, interval = "confidence")
```

```
##          fit          lwr          upr  
## 1 5.851413 5.786908 5.915918
```

```
#PI  
predict(model_final, newdata = predicted_data, interval = "prediction")
```

```
##          fit          lwr          upr  
## 1 5.851413 5.417609 6.285217
```

### 3.4 Summarize

After using backward AIC and VIF, we can conclude that the final model  $\text{price} = \text{intercept} + b_1\text{depth} + b_2\text{clarity} + b_3x$  is the best multiple Linear Regression Model to predict price. Based on CI and PI estimation on a new data created by us: clarity is SI2, depth is 60 mm, and width is 4 mm; We have confident interval is between [5.844164, 5.969452] and prediction interval is between [5.473483, 6.340133].